

API Spec: Export & Reporting API

TL;DR

A backend service for on-demand generation of Excel (.xlsx) exports from the scored request queue: (1) Open Requests Export—a concise, meeting-ready file listing only open requests (date_complete IS NULL) by descending Effort-Adjusted ROI Score, with seven core columns and ISO-date auto-naming (e.g., Open_Requests_2026-04-05.xlsx). (2) Full Queue Export—a detailed export of all (open and closed) requests, including granular score details for leadership review. Both exports capture the exact queue state at download, including manual rank overrides, and are accessible to Data Manager and Team Lead roles without additional authentication.

Goals

Business Goals

- Deliver secure, real-time queue exports supporting status transparency without exposing sensitive scoring internals externally.
- Equip leadership with verifiable, auditable queue snapshots for decision-making and compliance.
- Guarantee that any manual rank overrides by data managers persist into exported data.
- Increase operational efficiency by minimizing the manual work typically required for meeting prep and reporting.

User Goals

- Enable a one-click download of ranked open requests for immediate meeting use.
- Allow team leads and data managers to export the full queue (with all scores and detailed metadata) for offline analysis and reporting.
- Ensure exported files require no post-processing and are compatible with Excel/Google Sheets.
- Ensure exported rank order always matches what is currently displayed (including manual overrides).

Non-Goals

- No scheduled, automated exports—exporting is always user-triggered and on-demand.
 - No inclusion of PI Profile data or enrichment cache exports; scope is strictly the request queue and score outputs.
 - No emailing or alternate file delivery mechanisms; files are directly downloaded through the browser.
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User Stories

Personas: Data Manager, Team Lead

- As a **Data Manager**, I want to click a single button on the dashboard to instantly download a ranked Excel file of open requests, so I can prep for my team meeting with no manual formatting.
 - As a **Data Manager**, I want exported ranks to reflect any manual overrides I've applied, so exports always match the visible system.
 - As a **Team Lead**, I want to export a full queue with all score columns (Weighted Return Score, Effort-Adjusted ROI Score, and components) to support leadership reporting and offline review.
 - As a **Team Lead**, I want each export's filename to automatically include today's date, so repeated exports are always uniquely and clearly named for version control.
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Functional Requirements

Open Requests Export (Priority: High)

- **Trigger:** User clicks "Export Open Requests" on the ranked queue dashboard (Data Manager and Team Lead roles).
- **Filter:** Include only requests where `date_complete IS NULL` from REDCap/application data.
- **Sort:** Descending by Effort-Adjusted ROI Score (matches on-screen queue).
- **Override Rank Awareness:** If a request's rank has a manual override, the export reflects the override-adjusted rank.
- **Columns (order and logic):**
 1. Rank (integer, 1-based, override-aware)
 2. PI Name (Last, First from REDCap)
 3. Department (from REDCap)
 4. Funding Amount (awarded if present, else proposed, else "Not specified"; plain number)
 5. Study Title (Project Title from REDCap)
 6. Request Type (from REDCap)
 7. Effort-Adjusted ROI Score (rounded to 2 decimals)
- **File format:** Excel .xlsx
- **Filename:** `Open_Requests_YYYY-MM-DD.xlsx` (current date, ISO 8601)
- **Delivery:** File generated in memory, streamed to browser; never saved to server disk.

Full Queue Export (Priority: High)

- **Trigger:** User clicks "Export Full Queue" on dashboard (Data Manager and Team Lead).
- **Filter:** No filter; include all requests.

- **Sort:** Descending by Effort-Adjusted ROI Score; closed requests visually indicated.
- **Override Rank Awareness:** As above; manual overrides dominate.
- **Columns (order and logic):**
 1. Rank (override-aware)
 2. Status (Open/Closed via date_complete null check)
 3. PI Name
 4. Department
 5. Date of Request
 6. Date Complete (blank if open)
 7. Funding Amount (see logic above)
 8. Study Title
 9. Request Type
 10. Weighted Return Score (rounded to 2 decimals)
 11. Effort-Adjusted ROI Score (rounded to 2 decimals)
 12. Publication Impact Score
 13. Journal Prestige Score
 14. Funding Activity Score
 15. Research Output Score
 16. Enrichment Status (COMPLETE/PARTIAL/FAILED)
 17. Last Enrichment Date
- **File format:** Excel .xlsx
- **Filename:** Full_Queue_Export_YYYY-MM-DD.xlsx
- **Delivery:** In-memory, streamed to browser.

Export State Snapshot (Priority: High)

- Exports reflect the queue's exact state at generation: latest scoring per request, all current manual overrides, open/closed status.
- Reads from latest `request_scores`, does not trigger or recalculate scoring.
- Requests without a valid score (enrichment failure) appear with null scores in Full Queue Export and are excluded from Open Requests Export.

Rank Column Computation (Priority: High)

- Ranks calculated at export as:
 - Filter as appropriate (open/closed)
 - Sort by Effort-Adjusted ROI Score descending
 - Assign ranks 1,2,3...
 - Apply manual overrides: overridden ranks are inserted and other ranks shifted. Multiple overrides handled deterministically.
- Requests with null Effort-Adjusted ROI Score (failed enrichment) are ranked last in Full Queue (by Date of Request asc.), omitted from Open Requests.

Funding Amount Field Handling (Priority: Medium)

- Source: REDCap awarded amount field, fallback to proposed, else "Not specified".
- Format: Plain number; actual Excel currency formatting applied.

Export Audit Logging (Priority: Medium)

- Every export action is logged with: user, export type, timestamp, record count, filename.
 - Log is visible to Team Leads in admin panel.
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User Experience

Entry Point & First-Time User Experience

- Users access both export buttons directly from the main ranked queue dashboard.
- 'Export Open Requests (.xlsx)' and 'Export Full Queue (.xlsx)' are visible and enabled for both Data Manager and Team Lead roles.
- No onboarding/tutorial is required; button labels are clear.

Core Experience

1. **Step 1:** User clicks the desired export button.
 - Button disables with a visible loading indicator ("Generating export...").
 - File generation occurs server-side—typically <3 sec (Open Requests) or <5 sec (Full Queue)).
2. **Step 2:** Upon completion, file auto-downloads to user's default location.
 - UI displays a toast: 'Open_Requests_2026-04-05.xlsx downloaded successfully.'
3. **Step 3:** User opens the file in Excel or Google Sheets.
 - No reformatting required: headers are bold, alternating row shading assists legibility, columns auto-fit, header row frozen.
 - Full Queue Export highlights closed requests (Status column: Closed rows shaded light grey).
 - Enrichment Status is color-coded (COMPLETE/green, PARTIAL/amber, FAILED/red).
4. **Step 4:** User distributes file to meeting, leadership, or for personal review.

Advanced Features & Edge Cases

- Duplicate export actions are blocked (button remains disabled while file is generating).
- If export generation fails (>10 sec or backend error), user sees clear error message with retry option.
- For very large exports (>500 rows), export is streamed in batches to avoid memory bottlenecks.
- Audit trail is updated even on export failure (with result status).

UI/UX Highlights

- High color contrast for all conditional formatting (accessible even for color-blind users).
 - Responsive layout for queue dashboard (export buttons prominent and not hidden on any screen size).
 - Excel files open cleanly in both Excel and Google Sheets—tested on latest versions.
 - All columns are clearly labeled; no cryptic codes.
 - No modal/confirmation dialog (low-risk, nearly instantaneous).
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Narrative

It's Monday morning and the data manager has just refreshed the research request queue. With a single click on 'Export Open Requests', a perfectly formatted Excel sheet with live-ranked requests downloads instantly—no complex exports or manual resorting required. He posts the file in the team chat, where everyone sees not only the request list, but the exact order (with recent manual bumps for urgent deadlines) displayed on the dashboard. After the meeting, the team lead needs a full accounting for department leadership, so she clicks 'Export Full Queue', downloads the complete, score-detailed queue, and forwards it—knowing every score and status has a clear provenance. No one has to ask when the export was run because the date is in the filename. With zero friction and zero confusion, everyone is aligned and meetings can focus on action, not admin.

Success Metrics

Technical Metrics

- Open Requests Export generates in <3 seconds (up to 200 open requests)
- Full Queue Export generates in <5 seconds (up to 500 total requests)
- 100% fidelity: manual override ranks are always reflected exactly as in the UI
- Zero unreported failures (all errors surfaced to user and logged)
- 100% audit log coverage (export, user, timestamp, record count, filename)

User-Centric Metrics

95% exports are opened with no reformatting (survey or spot-checks)

90% team leads report “very satisfied” with new export process within 3 months

- Adoption tracked by number of exports per user per week (increase over previous manual period)

Business Metrics

- Meeting prep time for status reviews drops by 80% (self-reported by data managers)

98% of leadership reports are supplied directly from exports, not manual screenshots or ad-hoc lists

Technical Metrics

- Export generation times meet SLA 99%+ of the time
- <1% error rate in auditable export attempts
- Zero exports delivered with incorrect or missing override ranks

Tracking Plan

- User export button clicks (by export type)
 - File generated (filename, timestamp, record count)
 - Audit log entry created (user, result=success/fail)
 - Export generation time (alert if >10s)
 - Frequency of exports per user per week (adoption metric)
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Technical Considerations

Technical Needs

- Server-side Excel (.xlsx) generation using openpyxl, ExcelJS, or equivalent
- Reads from request_scores (latest scoring_version), REDCap request data, override_log
- Ranks calculated in-memory at export, not persistently stored
- Streaming file download direct to browser, never persistent on server

Integration Points

- Application database (REDCap tables, scoring tables, override_log)
- Audit trail logging (existing admin panel)
- No external APIs required for export generation

Data Storage & Privacy

- No files stored server-side; all data generated transiently in RAM.
- Audit log entries include user, export type, timestamp, and file name.
- Compliance with institutional data privacy by never exporting unpublished enrichment cache or PI Profile data.

Scalability & Performance

- Expected: up to 200 open requests (Open Export), up to 500 requests (Full Export)
- Large exports stream rows, minimize in-memory footprint
- Monitoring and alerting if export >10 seconds

Potential Challenges

- Edge-case handling for simultaneous/multiple override ranks
 - Funding Amount field names in REDCap may change—must be validated before release
 - Rare but possible: Export generation slows for >500 requests—address via pagination/streaming
 - Must handle interrupted exports gracefully (UI feedback, audit log entry even on fail)
 - UI must prevent double-submission (button disabled during export)
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Milestones & Sequencing

Project Estimate

- **Small:** 2–3 days total

Team Size & Composition

- **Small Team:** 1 backend developer, 1 frontend developer

Suggested Phases

Phase 1: Open Requests Export (Day 1)

- Key Deliverables: Backend—Open Requests Excel generation, override rank computation, in-memory streaming; Frontend—button placement, loading state.
- Dependencies: Confirm REDCap funding amount field mappings; override_log schema in place.

Phase 2: Full Queue Export & UI Polish (Day 1-2)

- Key Deliverables: Backend—Full export with all score columns, Status/enrichment color formatting; Frontend—button, loading, toast notifications.
- Dependencies: Completion of Phase 1.

Phase 3: Export Audit Logging & QA (Day 2-3)

- Key Deliverables: Backend—Audit log entries for each export; QA—verify override/score fidelity, error handling, Excel compatibility.
 - Dependencies: Backend export functions stable and UI connected.
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